

1.

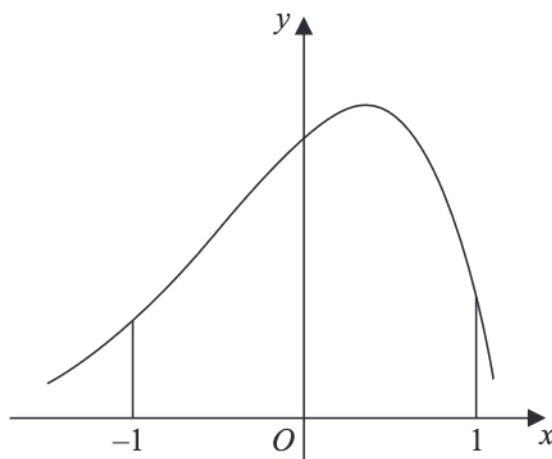


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$

The table below shows some corresponding values of x and y for this curve.

The values of y are given to 3 decimal places.

x	-1	-0.5	0	0.5	1
y	2.287	4.470	6.719	7.291	2.834

Using the trapezium rule with all the values of y in the given table,

(a) obtain an estimate for

$$\int_{-1}^1 f(x) \, dx$$

giving your answer to 2 decimal places.

(3)

(b) Use your answer to part (a) to estimate

(i) $\int_{-1}^1 (f(x) - 2) \, dx$

(ii) $\int_1^3 f(x-2) \, dx$

(3)



2.

In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

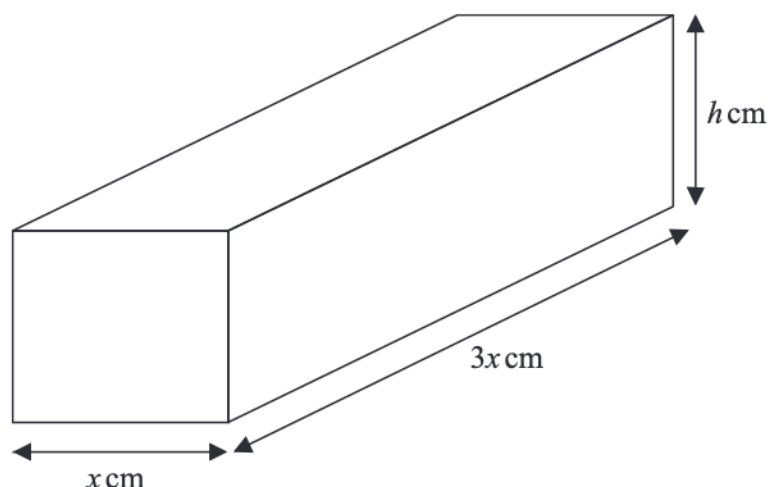


Figure 2

A brick is in the shape of a cuboid with width x cm, length $3x$ cm and height h cm, as shown in Figure 2.

The volume of the brick is 972 cm^3

(a) Show that the surface area of the brick, $S \text{ cm}^2$, is given by

$$S = 6x^2 + \frac{2592}{x} \quad (3)$$

(b) Find $\frac{dS}{dx}$ (1)

(c) Hence find the value of x for which S is stationary. (2)

(d) Find $\frac{d^2S}{dx^2}$ and hence show that the value of x found in part (c) gives the minimum value of S . (2)

(e) Hence find the minimum surface area of the brick. (1)



3. $f(x) = \left(2 + \frac{kx}{8}\right)^7$ where k is a non-zero constant

- (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of $f(x)$.
Give each term in simplest form.

(4)

Given that, in the binomial expansion of $f(x)$, the coefficients of x , x^2 and x^3 are the first 3 terms of an arithmetic progression,

- (b) find, using algebra, the possible values of k .

(Solutions relying entirely on calculator technology are not acceptable.)

(3)



4. (i) Using the laws of logarithms, solve

$$\log_3(4x) + 2 = \log_3(5x + 7) \quad (3)$$

- (ii) Given that

$$\sum_{r=1}^2 \log_a(y^r) = \sum_{r=1}^2 (\log_a y)^r \quad y > 1, a > 1, y \neq a$$

find y in terms of a , giving your answer in simplest form. (3)



5.

$$f(x) = x^3 + (p + 3)x^2 - x + q$$

where p and q are constants and $p > 0$

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that

$$9p + q = -51 \quad (2)$$

Given also that when $f(x)$ is divided by $(x + p)$ the remainder is 9

(b) show that

$$3p^2 + p + q - 9 = 0 \quad (2)$$

(c) Hence find the value of p and the value of q . (3)

(d) Hence find a quadratic expression $g(x)$ such that

$$f(x) = (x - 3)g(x) \tag{2}$$



6. The circle C has equation

$$x^2 + y^2 + 8x - 4y = 0$$

(a) Find

- the coordinates of the centre of C ,
- the exact radius of C .

(3)

The point P lies on C .

Given that the tangent to C at P has equation $x + 2y + 10 = 0$

(b) find the coordinates of P

(4)

(c) Find the equation of the normal to C at P , giving your answer in the form $y = mx + c$ where m and c are integers to be found.

(3)



7. A geometric sequence has first term a and common ratio r , where $r > 0$

Given that

- the 3rd term is 20
- the 5th term is 12.8

(a) show that $r = 0.8$

(1)

(b) Hence find the value of a .

(2)

Given that the sum of the first n terms of this sequence is greater than 156

(c) find the smallest possible value of n .

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(4)



8.

In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

- (i) Solve, for $-\frac{\pi}{2} < x < \pi$, the equation

$$5 \sin(3x + 0.1) + 2 = 0$$

giving your answers, **in radians**, to 2 decimal places.

(4)

- (ii) Solve, for $0 < \theta < 360^\circ$, the equation

$$2 \tan \theta \sin \theta = 5 + \cos \theta$$

giving your answers, **in degrees**, to one decimal place.

(5)



9.

In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

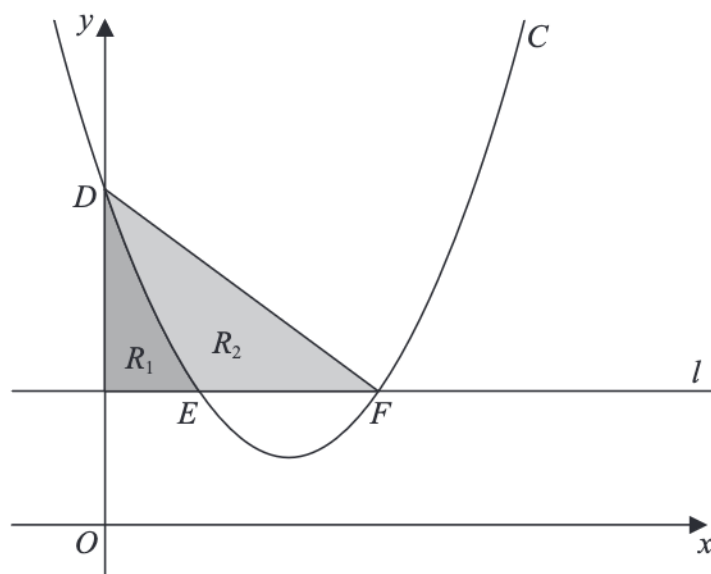


Figure 3

Figure 3 shows

- the curve C with equation $y = x^2 - 4x + 5$
- the line l with equation $y = 2$

The curve C intersects the y -axis at the point D .

- (a) Write down the coordinates of
- D
- .

(1)

The curve C intersects the line l at the points E and F , as shown in Figure 3.

- (b) Find the
- x
- coordinate of
- E
- and the
- x
- coordinate of
- F
- .

(2)

Shown shaded in Figure 3 is

- the region R_1 which is bounded by C , l and the y -axis
- the region R_2 which is bounded by C and the line segments EF and DF

Given that $\frac{\text{area of } R_1}{\text{area of } R_2} = k$, where k is a constant,

- (c) use algebraic integration to find the exact value of
- k
- , giving your answer as a simplified fraction.

(5)



10. A student was asked to prove by exhaustion that

if n is an integer then $2n^2 + n + 1$ is **not** divisible by 3

The start of the student's proof is shown in the box below.

Consider the case when $n = 3k$

$$2n^2 + n + 1 = 18k^2 + 3k + 1 = 3(6k^2 + k) + 1$$

which is not divisible by 3

Complete this proof.

(4)

